

Renal cell carcinoma

Two forks decide everything — **clear cell vs non-clear cell** histology, and **IMDC risk**. Localized disease is surgical; metastatic clear cell is an **IO backbone** (IO/IO or IO/TKI) with essentially no predictive biomarker

Stanford Oncology Handbook 2025, GU: Renal Cell Carcinoma · NCCN Kidney Cancer · items marked † are outside the handbook chapter

† WORKUP

- **CT abdomen/pelvis with contrast** (or MRI) defines T stage and venous extension; chest CT for staging. Brain/ bone imaging only if symptomatic
- **Renal mass biopsy is not required before nephrectomy** — get it for small masses under surveillance/ablation, suspected lymphoma or metastasis, or before systemic therapy without surgery
- Labs feeding IMDC: **CBC (Hgb, neutrophils, platelets), calcium**, LDH, creatinine, LFTs
- Grade (WHO/ISUP) is nucleolar: **G1** nucleoli inconspicuous at 400× · **G2** conspicuous at 400× · **G3** conspicuous at 100× · **G4** pleomorphism, giant cells, **rhaddoid or sarcomatoid**

GENETICS REFERRAL (NCCN)

- RCC diagnosed at **age <46**
- **Bilateral or multifocal** tumors
- **≥1 first- or second-degree relative** with RCC
- Somatic NGS finds a gene with germline implications (BRCA1/2, Lynch, etc.)

STAGING (AJCC)

T1	Confined to kidney <7 cm — T1a <4 cm, T1b 4–7 cm
T2	Confined to kidney >7 cm — T2a 7–10 cm, T2b >10 cm
T3	Into major veins or perinephric tissue, not beyond Gerota's — a renal vein / pelvicalyceal / sinus fat · b IVC below diaphragm · c IVC above diaphragm or caval wall
T4	Beyond Gerota's fascia, incl. contiguous ipsilateral adrenal
Stage	I T1N0 · II T2N0 · III T1–2N1 or T3 any N · IV T4 or any M1

IMDC RISK — 6 FACTORS

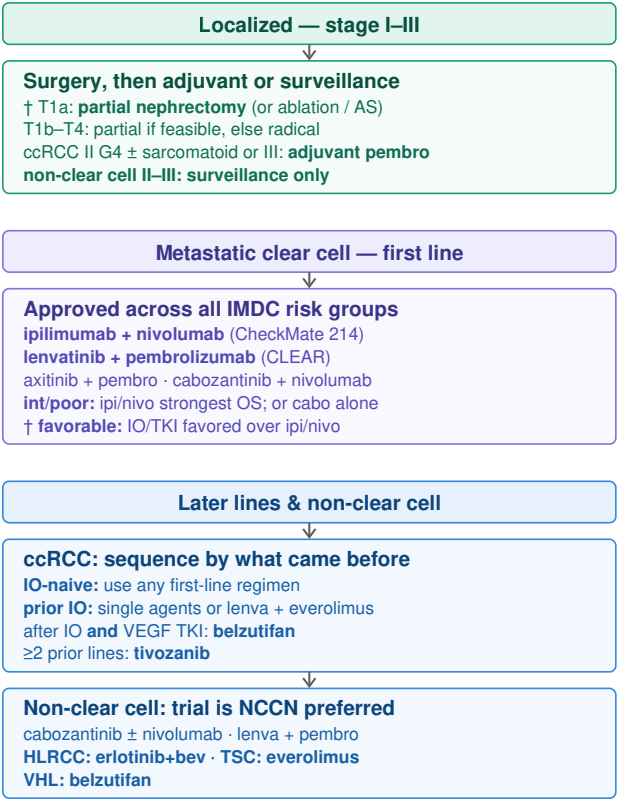
KPS <80	Hemoglobin < LLN
Dx → treatment <1 year	Calcium > ULN
Neutrophils > ULN	Platelets > ULN

Favorable = 0 · Intermediate = 1–2 · Poor = ≥3. Three of the six are simple counts — neutrophils, platelets, hemoglobin.

† PEARLS

- **Cytoreductive nephrectomy is no longer routine** — CARMENA showed sunitinib alone non-inferior to nephrectomy + sunitinib in int/poor risk. Reserve it for good-performance patients with limited metastatic burden and a large symptomatic primary
- **Sarcomatoid features** carry a poor prognosis but are the one histologic feature that predicts **unusually good response to IO** — favor ipi/nivo
- **Bone and brain metastases are common**; TKIs do not treat CNS disease — image the brain before starting therapy in symptomatic patients

TREATMENT PATHWAY



No predictive biomarker. PD-L1, TMB and MSI do not select therapy — the only biomarker-directed choices are syndromic. **CABOSUN and PAPMET are randomized phase 2**, not phase 3.

ADJUVANT AND FIRST-LINE CLEAR CELL

Regimen	Dosing	Cycle
Pembrolizumab (adjuvant)	300 mg or 600 mg IV	q3wk / q6wk
Sunitinib (adjuvant)	50 mg PO days 1–28	q42d
Ipilimumab + nivolumab	Ipi 1 mg/kg IV ×4; nivo 3 mg/kg ×4, then 240 or 480 mg	q3wk ×4, then q2/ q4wk
Lenvatinib + pembrolizumab	Lenvatinib 20 mg PO daily; pembro 200 or 400 mg IV	q3wk / q6wk
Axitinib + pembrolizumab	Axitinib 5 mg PO BID; pembro 200 or 400 mg IV	q3wk / q6wk
Cabozantinib + nivolumab	Cabozantinib 40 mg PO daily; nivo 240 or 480 mg IV	q2wk / q4wk
Axitinib + avelumab	Axitinib 5 mg PO BID; avelumab 10 mg/kg IV	q2wk
Cabozantinib single agent	60 mg PO daily	continuous
Pazopanib · sunitinib	800 mg PO daily · 50 mg PO days 1–28	daily · q42d

IO is capped at **24 months** across these regimens. Historical: temsirolimus 25 mg IV weekly (ARCC) and high-dose IL-2 600,000 IU/kg IV q8h days 1–5 and 15–19, inpatient, ORR 14%.

LATER LINES AND NON-CLEAR CELL

Regimen	Dosing	Cycle
Lenvatinib + everolimus	Lenvatinib 18 mg + everolimus 5 mg PO daily	continuous
Belzutifan	120 mg PO daily — after IO and VEGF TKI	continuous
Tivozanib	1.34 mg PO — 3 wk on, 1 wk off, after ≥2 lines	q4wk
Axitinib	5 mg PO BID → 7 mg BID if BP ≤150/90 , max 10 mg BID	continuous
Erlotinib + bevacizumab	Erlotinib 150 mg PO daily; bev 10 mg/kg IV — FH-deficient / HLRCC	q2wk

LANDMARK TRIALS

Trial	Result
KEYNOTE-564	Adjuvant pembro, high-risk ccRCC — 48-mo DFS 64.9% vs 56.6% , OS 91.2% vs 86% (HR 0.72)
CheckMate 214	Ipi/nivo vs sunitinib, int/poor — OS 47.0 vs 26.6 mo , CR 11% vs 2%; no benefit in favorable risk
CLEAR	Lenva/pembro — PFS 23.9 vs 9.2 mo , ORR 71.3% vs 36.7% (longest 1L PFS reported)
KEYNOTE-426 · CheckMate 9ER	Axi/pembro OS 46 vs 40 mo · cabo/nivo OS 46.5 vs 36 mo, both vs sunitinib
LITESPARK-005	Belzutifan after IO + TKI — ORR 21.9% vs 3.5%; median PFS 5.6 mo in both arms; OS failed
TIVO-3	Tivozanib after ≥2 lines — PFS 5.6 vs 3.9 mo, no OS benefit
SWOG 1500 (PAPMET)	Papillary — cabozantinib PFS 9.0 vs 5.6 mo; MET inhibitors failed
S-TRAC	Adjuvant sunitinib — DFS 6.8 vs 5.6 yr, no OS benefit

† **Surveillance** is not in the chapter — NCCN: exam, labs and cross-sectional imaging q3–6 mo ×3 yr after resection of stage II–III, then annually to year 5.